Quantification: 877_wt_z081c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	8
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3075.75	2935.5	2349.76	px^2
major_axis_length_px	lacuna shape	88.2344	96.7141	43.2425	px
minor_axis_length_px	lacuna shape	43.1291	45.335	17.1087	px
aspect_ratio	lacuna shape	1.9916	1.9763	0.4765	unitless
eccentricity	lacuna shape	0.8367	0.8626	0.0884	unitless
solidity	lacuna shape	0.7992	0.8371	0.1049	unitless
roots_count	headline	11.375	10	6.3005	count
ring_length_r30_px	headline	425.875	392	212.754	px
ring_length_r60_px	reported	1022.62	1045	423.982	px
owned_length_px	ownership-dependent	2544.14	2512.29	1523.3	px
edge_count	ownership-dependent	122.875	121.5	73.1602	count
mean_edge_length_px	ownership-dependent	20.8461	20.4096	2.0328	px
perimeter_px	normalised	277.935	293.72	141.482	px
ring_area_r30_px2	normalised	9306.25	9005.5	4078.1	px^2
ring_area_r60_px2	normalised	22657	22110	8631.97	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.045787	0.0500705	0.0134898	px^-1
ring_density_r60	normalised	0.046033	0.048176	0.0140076	px^-1
roots_per_100px_perimeter	normalised	4.2514	3.5955	1.4209	per 100 px
ring_attached_length_r30_px	network v2	382.625	361.5	226.634	px
ring_attached_length_r60_px	network v2	848.875	878.5	459.247	px
ring_length_w_r30_px	network v2	487.971	444.94	247.515	px
ring_length_w_r60_px	network v2	1178.47	1209.49	491.525	px
sholl_crossings_r10	network v2	10.125	8.5	6.7069	count
sholl_crossings_r20	network v2	12.625	13	5.8782	count
sholl_crossings_r30	network v2	14.375	15	6.9885	count
loop_count	loops, report only	10.5	11	6.8868	count
loop_edge_share	loops, report only	0.322	0.3261	0.1938	share
ring_width_mean_r30_px	width	6.7157	6.7224	0.956	px

Per field
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measure	value	units
analysed_area_px2	1.0135e+06	px^2
total_skeleton_length_px	33783	px
canalicular_length_density_per_px	0.0333332	px^-1
junction_count	719	count
junction_density_per_px2	0.00070943	px^-2
lacuna_count	12	count

lacunae_per_px2	1.184e-05	px^-2
skeleton_component_count	324	count
mean_component_length_px	104.269	px
median_component_length_px	24.5	px
field_length_density_w_per_px	0.0381538	px^-1
field_density_without_flagged_per_px	0.0333332	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	461	px
unattached_component_count	28	count
canalicular_length_density_attached_per_px	0.0333332	px^-1
field_length_density_with_unattached_per_px	0.033788	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	111	count
loop_edge_share	0.2681	share
edge_count	1712	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4977	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	33783	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	828	47.9341	26.2785	1.8241	0.8363	0.811	1.1987	10.5652	544.853	True
2	4064	125.912	44.1728	2.8504	0.9364	0.8472	0.9481	38.4616	829.728	True
3	6500	135.386	68.4723	1.9772	0.8627	0.7101	0.7764	202.316	509.649	False
4	493	36.8679	18.6636	1.9754	0.8624	0.8203	-0.7656	201.503	648.073	False
5	4498	128.704	46.6689	2.7578	0.9319	0.9623	0.3797	397.582	25.7919	True
6	4895	134.554	48.0863	2.7982	0.934	0.8829	0.4584	399.457	314.021	False
7	1388	60.0833	43.9829	1.3661	0.6813	0.6761	-0.1762	427.803	598.192	False
8	1690	77.7941	42.0098	1.8518	0.8417	0.6483	0.0159	450.566	636.056	False
9	5044	117.23	57.6341	2.034	0.8708	0.8539	1.376	449.833	814.625	False
10	1085	58.3256	24.5501	2.3758	0.9071	0.9313	0.4295	634.654	16.1935	True
11	4181	115.634	46.6872	2.4768	0.9149	0.8959	1.0406	724.949	827.982	False
12	415	28.3253	19.4963	1.4529	0.7254	0.9061	-0.3467	864.431	580.06	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	223	624	1468.86	69	21.2879	7.0578
2	10	340	833	1069.56	45	23.768	6.7986
3	23	809	1670	5078.3	238	21.3374	7.1131
4	8	347	974	1665.85	90	18.5095	8.0879
5	6	247	711	1132	46	24.6088	8.1573

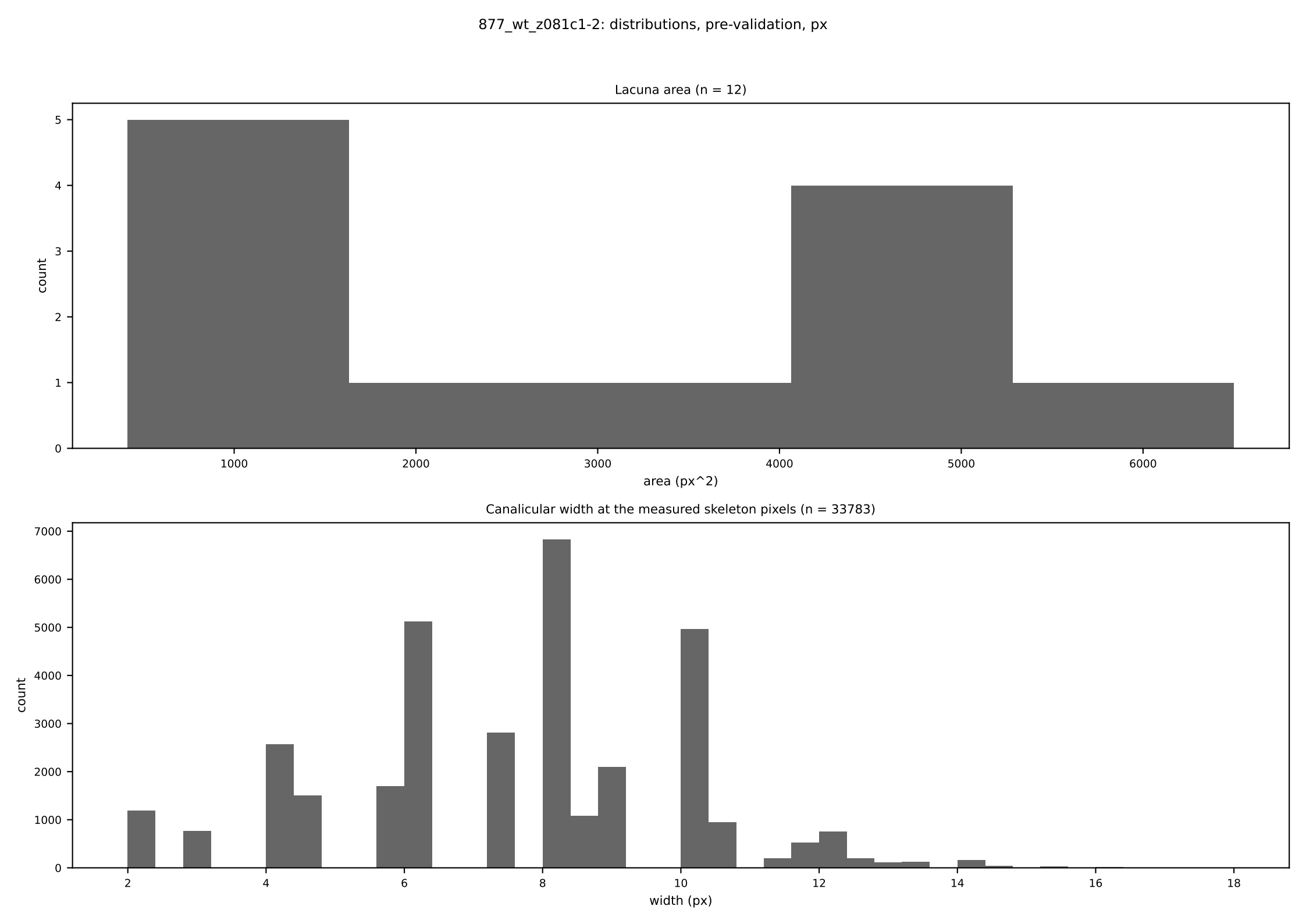
6	11	437	1182	3040.54	151	20.1361	7.7222
7	8	304	699	1570.13	63	24.9227	5.6775
8	9	335	856	1984.03	103	19.2624	6.3256
9	18	609	1392	3749	191	19.6283	7.0826
10	4	124	375	677.914	22	30.8143	6.1111
11	11	465	1116	3120.51	140	22.2894	6.3622
12	3	101	292	144.782	7	20.6831	5.3541

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	149.397	4144	11016	0.5976	0.5673	0.0538127	0.0566449	6.0242
2	351.404	9408	21117	0.7903	0.7218	0.0361395	0.0394469	2.8457
3	515.671	15609	34048	1	1	0.0518291	0.0490484	4.4602
4	108.397	5713	15341	1	1	0.0607387	0.06349	7.3803
5	326.735	7997	18643	0.6576	0.6224	0.0308866	0.0381376	1.8364
6	354.617	12438	30402	1	1	0.0351343	0.038879	3.1019
7	216.108	5237	12563	1	1	0.0580485	0.0556396	3.7019
8	263.451	6630	15255	1	1	0.0505279	0.0561128	3.4162
9	355.262	12275	29427	1	1	0.049613	0.0473035	5.0667
10	148.953	4761	12108	0.6788	0.6158	0.026045	0.0309713	2.6854
11	323.99	11381	28242	1	1	0.0408576	0.0395156	3.3952
12	85.9828	5167	15978	1	1	0.0195471	0.0182751	3.4891

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	223	570	262.522	727.169	6	8	7
2	309	747	390.363	962.891	10	10	11
3	808	1648	938.092	1936.57	23	21	26
4	344	974	393.22	1113.89	5	9	13
5	205	585	279.38	817.896	6	9	11
6	379	987	496.659	1357.57	11	15	16
7	185	528	355.262	814.151	6	11	9
8	280	674	378.421	974.767	4	10	14
9	567	1118	702.441	1594.14	14	17	17
10	81	190	142.125	431.233	3	4	5
11	419	783	526.032	1305.09	14	16	18
12	79	79	113.64	331.605	4	2	2


877_wt_z081c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1000
2000
3000
4000
5000
6000
Canalicular width at the measured skeleton pixels (n = 33783)
count
width (px)
0
1000
2000
3000
4000
5000
6000
7000
2
4
6
8
10
12
14
16
18